

Module 4: Switching, Cellular Networks & LANs

Circuit vs Packet, Cellular Frequency Reuse, CSMA/CD, CSMA/CA, VLAN

Module IV: Switching, Cellular Networks & LANs — Topics Covered

1. Circuit Switching vs. Packet Switching
2. Cellular Frequency Reuse ($N = i^2 + ij + j^2$)
3. IEEE 802.3 Ethernet & CSMA/CD Backoff
4. IEEE 802.11 Wi-Fi & CSMA/CA (RTS/CTS)
5. Virtual LANs (VLANs & IEEE 802.1Q)

1. Ethernet CSMA/CD vs. Wi-Fi CSMA/CA

Feature	IEEE 802.3 Ethernet (CSMA/CD)	IEEE 802.11 Wi-Fi (CSMA/CA)
Mechanism	Carrier Sense Multiple Access with Collision Detection	Carrier Sense Multiple Access with Collision Avoidance
Wireless Challenge	Transceiver can detect collision while sending.	Cannot detect collision while transmitting (RF blindness) $\implies$ Must avoid collisions.
Collision Recovery	Emits 32-bit Jam signal; Binary Exponential Backoff ($[0, 2^k - 1] \times 51.2\mu s$).	Interframe Spaces (DIFS, SIFS); RTS/CTS handshaking resolves hidden terminals.